



Diesel Generator Set

mtu 12V4000 DS2250

380V – 11 kV/50 Hz/data center continuous power/
NEA (ORDE) + Tier 2 optimized/12V4000G34F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 2020 kVA - 2100 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 100% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions

- Tier 2 optimized engine
- NEA (ORDE) optimized engine

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)	
Manufacturer	mtu		Total oil system capacity: l	260
Model	12V4000G34F		Engine jacket water capacity: l	160
Type	4-cycle		Intercooler coolant capacity: l	40
Arrangement	12V		Combustion air requirements	
Displacement: l	57.2			
Bore: mm	170			
Stroke: mm	210		Combustion air volume: m³/s	2.2
Compression ratio	16.4		Max. air intake restriction: mbar	50
Rated speed: rpm	1500		Cooling/radiator system	
Engine governor	ADEC (ECU 9)			
Max power: kWm	1755			
Air cleaner	dry		Coolant flow rate (HT circuit): m³/hr	55
Fuel system			Coolant flow rate (LT circuit): m³/hr	30
			Heat rejection to coolant: kW	675
			Heat radiated to charge air cooling: kW	430
			Heat radiated to ambient: kW	75
			Fan power for electr. radiator (40°C): kW	55
Maximum fuel lift: m	5		Exhaust system	
Total fuel flow: l/min	27			
Fuel consumption ²⁾				
	l/hr	g/kwh	Exhaust gas temp. (after engine): °C	460
At 100% of power rating:	413	195	Exhaust gas temp., max (after engine): °C	550
At 75% of power rating:	324	204	Exhaust gas temp. (before turbocharger): °C	700
At 50% of power rating:	229	216	Exhaust gas volume: m³/s	5.5
			Maximum allowable back pressure: mbar	50

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NEA (ORDE) + Tier 2 optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 S7 (Low voltage Leroy Somer standard)	380 V	1680	2100	3191	1624	2030	3084
	400 V	1680	2100	3031	1624	2030	2930
	415 V	1680	2100	2922	1624	2030	2824
Leroy Somer LSA52.3 L12 (Low voltage Leroy Somer oversized)	380 V	1680	2100	3191	1624	2030	3084
	400 V	1680	2100	3031	1624	2030	2930
	415 V	1680	2100	2922	1624	2030	2824
Marathon 744RSL7092 (Low voltage Marathon)	380 V	1672	2090	3175	1616	2020	3069
	400 V	1672	2090	3017	1616	2020	2916
	415 V	1672	2090	2908	1616	2020	2810
Leroy Somer LSA53.2 XL9 (Medium volt. Leroy Somer)	11 kV	1680	2100	110	1632	2040	107
Marathon 1020FDH7097 (Medium volt. Marathon)	11 kV	1664	2080	109	1616	2020	106

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.